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A mechanistic critical heat flux model for wide range of subcooled and low quality flow boiling

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Abstract

A mechanistic model to predict a critical heat flux (CHF) over a wide operating range in the subcooled and low quality flow boiling has been proposed based on a concept of the bubble coalescence in the wall bubbly layer. The conservation equations of mass, energy and momentum, together with appropriate constitutive relations, are solved analytically to derive the CHF formula. The model is characterized by an introduction of the drag force due to wall-attached bubbles roughness in the momentum balance, which determines the limiting transverse interchange of mass flux crossing the interface of the wall bubbly layer and core. Comparison between the predictions by the proposed model and the experimental CHF data shows good agreement over a wide range of parameters for both light water and fusion reactors operating conditions. The model correctly accounts for the effects of flow variables such as pressure, mass flux and inlet subcooling as well as geometry parameters.

1. Introduction

The subject of critical heat flux (CHF) has received considerable attention over the past several decades, because the CHF in boiling phenomena presents one of the limitations in the design of boiling heat transfer system. However, unfortunately the knowledge is not sufficient to predict the CHF with the desired degree of accuracy. The complication of a CHF problem for forced convection boiling is caused by the large number of variable factors and the variety of two-phase flows. In order to understand the physical nature of the CHF phenomenon, various mechanistic CHF models have been proposed so far. According to recent reviews about the mechanistic modeling of CHF (Weisman, 1992; Katto, 1994; Celata, 1997), among many existing models available today, only near wall bubble crowding (Weisman and Pei, 1983; Ying and Weisman, 1986; Weisman and Ileslamlou, 1988) and liquid sublayer dryout (Lee and Mudawar, 1988; Katto, 1992; Celata et al., 1994a) models are receiving attention for the prediction of the CHF in the

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moderate and highly subcooled conditions. However, comparative analyses performed by Iwamura et al. (1992), Celata et al. (1994b) and Bricard and Souyri (1995) show that the predictive capabilities of all the existing mechanistic CHF models are not satisfactory enough to apply them to new system designs.

The near wall bubble crowding model was first developed by Weisman and Pei (1983). The model is based on the existence of a bubbly layer adjacent to the heated surface. It assumes that CHF occurs when the void fraction of the bubbly layer reaches a critical value, because bubble crowding prevents the bulk cold liquid from reaching the wall. Turbulent interchange between the bubbly layer and core is the limiting mechanism of the onset of CHF. While the liquid sublayer dryout model, first proposed by Lee and Mudawar (1988), is based on the existence of vapor slug adjacent to the tube wall. It assumes that CHF occurs at the dryout condition of the thin liquid layer beneath a vapor slug that is generated by Helmholtz instability at the interface between the liquid layer and vapor region. Both types of CHF models are capable of providing reasonably accurate predictions being very close when compared with each other.

Chang and Lee (1989) proposed a mechanistic CHF model based on the same concept of Weisman and Pei, while the CHF formula was derived from local balance equations. The limiting transverse mass flux crossing the interface of the bubbly layer and core was determined from a momentum balance. Lee et al. (1997) revised the original model with fixing mistakes included in the momentum balance, which were pointed out by Bricard and Souyri (1995) and they modified the model by introducing several new constitutive relations.

In this paper, some significant modifications are made on the physical concept of a CHF mechanism and relevant constitutive models in the bubble crowding model. A concept of single wall-attached bubbles layer replaces the bubbly layer in the previous models, composing of multiple bubbles in thickness. Wall-attached bubbles act as a surface roughness equivalent to bubble size. Because the reasoning of Chang and Lee's model, in the author's opinion, provides a sound starting point, the present model utilizes its analytical frame. The model is validated on the experimental CHF data of water and Freons in uniformly heated tubes over wide ranges of parameters and compared with the prediction capabilities of the CHF look-up table of Groeneveld et al. (1996).

2. Physical basis for the proposed model

Most of the existing mechanistic models are based on the hypothetical flow structure or the limited experimental observation at CHF condition. Attempted in this study is the development of a new prediction model of the flow boiling CHF based on the suggested wall-attached bubbles layer or wall bubbly layer. A physical image of the boiling structure considered in this study is shown in Fig. 1(a). The flow is divided into two regions. In the outer annular layer of the round tube, bubbles are packed on the wall just prior to agglomeration and located in the middle of the tube is a mixture core consisting of liquid and bubbles. The flow structure is based on the experimental observation by Gunther (1951) and Kirby et al. (1967) that the two-phase layer of local vapor film in subcooled flow boiling was an order of one bubble thickness.

According to Saha and Zuber (1974), at low mass flow rates, bubble detachment from the wall is thermally controlled and independent of hydrodynamic forces. While at high mass flow rates, bubbles do not easily detach from their nucleation sites because bubbles are small enough to prevent bubble detachment from the wall by the hydrodynamic forces. Therefore, the wall-attached bubbles form a layer, which acts as a wall roughness. Bubble departure depends on the drag force, which, in turn, depends upon the characteristic skin friction experienced by the wall-attached bubbles layer. Once the drag and buoyancy forces are strong enough to overcome the surface tension force which holds the bubbles to the wall, the bubbles tear free and move immediately into the core or slide along, the heated surface, then new bubbles are rapidly created at these sites. This point is called the onset of significant vapor generation (OSV).

Generally, active nucleation site density drastically increases when heat flux approaches to the CHF. As the number of bubbles surrounding one bubble becomes numerous, bubbles laterally coalesce and the voidage of wall bubbly layer increases. It is assumed that CHF condition reaches at a certain void fraction in the wall bubbly layer (called critical wall-void fraction) when radial thermal transport is limited by equal flows inward and outward at the interface of the wall bubbly layer and core. If the wall heat flux exceeds the maximum permissible heat flux satisfying the limiting condition above, the liquid in the wall bubbly layer depletes then dry spot area rapidly increases and consequently CHF occurs. It is assumed that a single layer of bubbles compactly attached on the wall play an effective physical barrier to the heat transfer from the wall and to the liquid supply from the core at near CHF condition.

Earlier work by Beattie (1980) has shown that wall bubbles act as a surface roughness equivalent to bubble size. Under such a condition, the friction factor could be described by the same form of equation as that used for a single-phase flow in roughened tubes. Although Beattie considered the flow remote from the CHF, it can be supposed that bubbles in the CHF region behave similarly. Therefore, the wall-attached bubbles are considered as increasing the wall roughness of the tube. In order to carry out above reasoning, it should be satisfied that the bubble diameter exceeds the thickness of the viscous sublayer. The existence of wall roughness changes the hydrodynamic characteristics of the flow and the effect of viscous shear due to molecular friction becomes relatively small.

The present model is based on the following postulates. (1) Under the subcooled and low quality flow boiling conditions, CHF is a local phenomenon. (2) At near CHF condition, bubbles are assumed to be packed in a most dense array in the wall bubbly layer, which has an effective thickness of a single bubble diameter. (3) CHF condition reaches at a certain critical wall-void fraction when radial thermal transport is limited by equal flows inward and outward from the outer edge of the wall bubbly layer. (4) The thermal transport limitation near the heated wall is governed by local mass, energy and momentum balance equations. The limiting transverse interchange of mass flux is mainly controlled by the convective shear effect due to the drag force on the wall-attached bubbles.

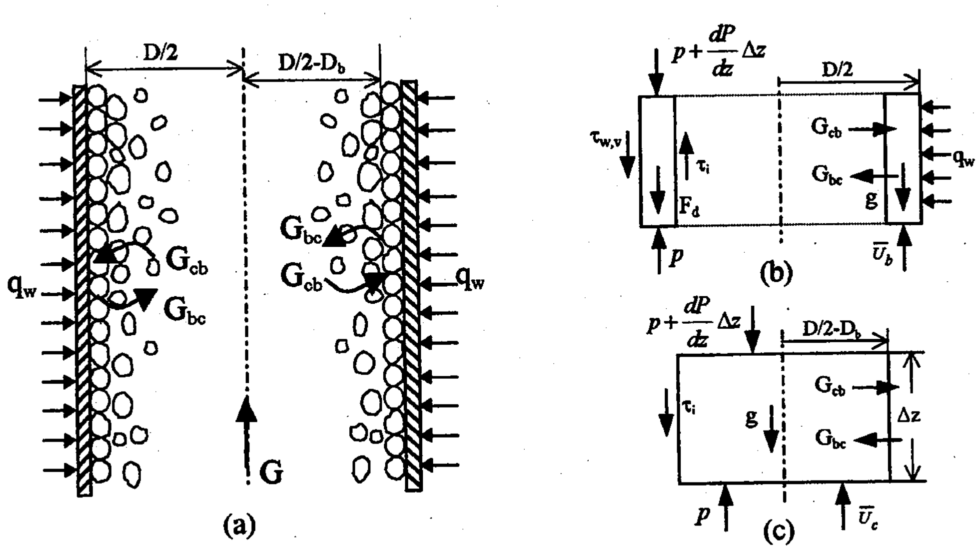

Fig. 1. (a) Conceptual configuration of bubbles on the heated wall and separated flow control volumes for (b) wall bubbly layer and (c) core.

3. The proposed CHF model

3.1. Conservation equations

In the present study, one-dimensional steady state subcooled and low quality flow boiling in a tube is analyzed with the effects of two-phase flow models and constitutive equations. Fig. 1(a) shows the transverse interchange crossing the interface of the wall bubbly layer and core. Because the transverse exchange phenomena are quite complicated and difficult to decompose into more elementary terms, it is modeled as the lumped term including turbulent mixing in bulk flow, vaporization, condensation and liquid agitation etc. The transverse mixing effect is very important and useful in the development of the present model.

The basic assumptions of the present model for mathematical formulation are listed below. (1) Two-phase flow structure at near CHF condition always takes the form of Fig. 1(a). (2) The flows in the wall bubbly layer and core are each homogeneous. (3) The flow is steady. (4) The axial change of pressure is negligible compared with the system pressure. (5) The wall bubbly layer thickness is very small compared to the tube radius.

The governing equations are derived by applying the basic conservation statements for mass, energy and momentum to the control volumes such as that shown in Fig. 1(b, c). Considering the bubbly layer control volume of Fig. 1(b), balances of total mass and liquid mass are given as below, respectively,

$$\frac{dG_bA_b}{dz} + G_{bc}\xi_i - G_{cb}\xi_i = 0 \tag{1}$$

$$\frac{dG_b(1-x_b)A_b}{dz} + \frac{q_{evap}\xi_w}{h_{fg}} + G_{bc}(1-x_b)\xi_i - G_{cb}(1-x_c)\xi_i = 0 \tag{2}$$

where G_{cb} and G_{bc} are the total inward and outward mass fluxes at the interface of the wall bubbly layer and core. ξ_i and ξ_w are the perimeters of the interface and the heated wall, respectively, which are expressed by $\xi_i = \pi(D-2D_b)$ and $\xi_w = \pi D$. Also q_{evap} represents the heat flux required to form vapor in the wall bubbly layer. Since the wall void fraction α_b is limited at the CHF condition, the quality in the wall bubbly layer has a finite value, i.e. $dx_b/dz = 0$, with the assumption of homogeneous flow in the bubbly layer. The variation of saturation properties of liquid and vapor in the axial flow direction is neglected because the pressure drop along the tube is small compared with the system pressure. Then two mass balance equations of (1) and (2) can be combined as below.

$$q_{evap} = G_{cb}(x_b - x_c)h_{fg}\frac{\xi_i}{\xi_w} \tag{3}$$

A total energy balance on the bubbly layer control volume yields

$$\frac{dG_bA_bh_b}{dz} + G_{bc}h_b\xi_i - G_{cb}h_c\xi_i - q_w\xi_w = 0 \tag{4}$$

Using the relationship of Eqs. (1) and (17) with the same assumptions above, Eq. (4) is rearranged as

$$q_w = G_{cb}(h_b - h_c)\frac{\xi_i}{\xi_w} \tag{5}$$

By introducing the factor of F_q, Eqs. (3) and (5) can be combined into

$$q_w = q_{evap}/F_q \tag{6}$$

with

$$F_q = \frac{(x_b - x_c)h_{fg}}{(h_b - h_c)} \tag{7}$$

where F_q represents the fraction of the heat flux producing vapor that enters the core region. The qualities appeared in the above equations are the actual flow qualities based on a nonequilibrium condition and x_b is the quality corresponding to the critical wall-void fraction α_b. It should be noted that the general approach to determine the boiling heat flux by using of mass and energy balances on the bubbly layer is the same as done by Weisman and Ileslamlou (1988).

Based on the one-dimensional momentum equation of a separated flow model, momentum balances between shear force, drag force and pressure on the control volumes of the wall bubbly layer and core are written, respectively, as

$$-\frac{dP}{dz}+\frac{\tau_i\xi_i}{A_b}-\frac{\beta\tau_{w,v}\xi_w}{A_b}-\frac{F_dN_{bub}}{A_bdz}-\rho_bg+\frac{(G_{ch}\bar{U}_c-G_{bc}\bar{U}_b)\xi_i}{A_b}=\frac{1}{A_b}\frac{d}{dz}(\rho_b\bar{U}_b^2A_b) \tag{8}$$

$$-\frac{dP}{dz}-\frac{\tau_i\xi_i}{A_c}-\rho_cg+\frac{(G_{bc}\bar{U}_b-G_{ch}\bar{U}_c)\xi_i}{A_c}=\frac{1}{A_c}\frac{d}{dz}(\rho_c\bar{U}_c^2A_c) \tag{9}$$

As in Taylor et al. (1985), the apparent wall shear stress τ_w is defined as the sum of the viscous shear $\tau_{w,v}$ and form drag force on the wall in the mean flow direction, which approach is called the discrete element method. The parameter F_d is the drag force which the rough element of a single wall-attached bubble exerts on the flow field and $\beta(0<\beta<1)$ is the blockage factor that represents the area of the heated surface between the roughness elements of bubbles. The number of attached bubbles along Δz in the wall bubbly layer is $N_{bub}=D\,dz/D_b^2$ when bubbles are arranged in a rectangular lattice with a pitch of D_b.

Subtracting Eq. (8) from Eq. (9) to eliminate the pressure gradient term in the left-hand side, we obtain the limiting transverse interchange of mass flux at the interface of the wall bubbly layer and core

$$G^*=\left[\frac{-\tau_i\xi_i}{\eta_c(1-\eta_c)A}+\frac{\beta\tau_{w,v}\xi_w}{(1-\eta_c)A}-(\rho_c-\rho_b)g+\frac{\pi DF_d}{D_b^2(1-\eta_c)A}+\Phi_{acc}\right]\frac{A\eta_c(1-\eta_c)}{(\bar{U}_c-\bar{U}_b)\xi_i} \tag{10}$$

with the definitions of the fraction of cross-sectional area occupied by the core and the acceleration term, respectively.

$$\eta_c=\frac{A_c}{A}=\frac{(D-2D_b)^2}{D^2} \tag{11}$$

$$\Phi_{acc}=\frac{1}{A_b}\frac{d}{dz}(\rho_b\bar{U}_b^2A_b)-\frac{1}{A_c}\frac{d}{dz}(\rho_c\bar{U}_c^2A_c) \tag{12}$$

It is considered here that the wall-attached bubbles act as surface roughness elements hydro-dynamically and the element size is represented by the characteristic length of the bubble departure diameter, which will be discussed in detail later. At CHF condition, the transverse mass transport rate at the interface is limited, i.e. $G^*=G_{bc}=G_{cb}$, based on the assumption in Section 2. Because the bubble departure diameter is much smaller than the tube radius, variation of shear stress across the thin wall bubbly layer can be neglected, i.e. $|\tau_i|\cong|\tau_{w,v}|$. If we assume that the bubbly layer behaves like the boundary layer region in a single-phase pipe flow, then the wall shear stress $\tau_{w,v}$ can be calculated from the Newton's viscous law using the universal velocity profiles. According to the rough calculation, the effects of shear stresses are negligible compared to the other terms in the bracket of Eq. (10) for both cases of $\beta=0$ and 1. Therefore, for simplicity in this calculation, the first and second shear stress terms in the bracket are disregarded. This is consistent with the hypothesis that the wall-attached bubbles play an important role on the CHF. This approach may seem crude, but the results are not strongly affected by the simplified assumption.

Chang and Lee (1989) have shown that the acceleration term Φ_{acc} is also negligibly small with respect to the radial mixing flow effect. Theoretically, Φ_{acc} can be calculated using two-phase flow identities, but it brings in more complexity without improvement of accuracy in CHF prediction. The acceleration influence is also neglected for a simple calculation in the present study. Finally the CHF formula, for given geometry and inlet fluid conditions, is expressed from Eqs. (5) and (10) in a simple form.

$$q_{CHF}=\left[-(\rho_c-\rho_b)g+\frac{\pi DF_d}{D_b^2(1-\eta_c)A}\right]\times\frac{(h_b-h_c)A\eta_c(1-\eta_c)}{(\bar{U}_c-\bar{U}_b)\xi_w} \tag{13}$$

In the CHF formula of Eq. (13), the enthalpy transportation between the wall bubbly layer and core through the interface is a critical value. An insight into the relationship between the CHF formula and the various parameters appearing in the above equations can be drawn. It indicates that the local drag force on the wall-attached bubbles and local flow enthalpy as well as mass flux, tube geometry and system pressure are major parameters. This implies that both hydrodynamic and thermal factors play a role in the onset of CHF as pointed out by Tong and Hewitt (1972).

3.2. Other relations

3.2.1. Two-phase flow properties

As shown in Eq. (13), accurate predictions of flow enthalpy and density at the location of the CHF occurrence are very important. In the case of uniform heating, the CHF occurrence is usually located in the downstream end of the heated tube. Assuming homogeneous flows of the wall bubbly layer and core, the quantities at the tube exit are defined as below

$$\alpha_c = \frac{1}{\eta_c}\alpha_{avg} - \frac{1-\eta_c}{\eta_c}\alpha_b \tag{14}$$

$$\rho_b = \rho_f(1-\alpha_b) + \rho_g\alpha_b \tag{15}$$

$$\rho_c = \rho_l(1-\alpha_c) + \rho_g\alpha_c \tag{16}$$

$$h_b = h_f(1-x_b) + h_g x_b \tag{17}$$

$$h_c = h_l(1-x_c) + h_g x_c \tag{18}$$

$$\mu_{2\phi} = \mu_f(1-\alpha_{avg})(1+25\alpha_{avg}) + \mu_g\alpha_{avg} \tag{19}$$

The average viscosity of core $\mu_{2\phi}$ is evaluated by Beattie and Whalley (1982) and the average bulk void fraction α_{avg} is determined by Eq. (40). Physical properties are calculated based on the tube exit pressure.

3.2.2. Thickness and critical void fraction of the wall bubbly layer

The bubbly layer thickness is a characteristic parameter in the bubble crowding model. It was assumed to be 2.5 times of the detached bubble diameter by Weisman and Pei (1983) and 1.7 times by Chang and Lee (1989). The thickness was determined empirically by best fitting the CHF model to a large number of data points. In the present study, however, the effective thickness of the wall bubbly layer is considered as a single bubble diameter. It is assumed that a single layer of wall bubbles provides sufficient barrier to the heat transfer from the wall and to the cold liquid supply from core, when bubbles are densely packed on the heated wall near the CHF condition. Detached bubbles could exist near the wall bubbly layer for low quality flow boiling. The existence of detached bubbles indirectly affects the CHF in terms of the thermo-physical properties in the core.

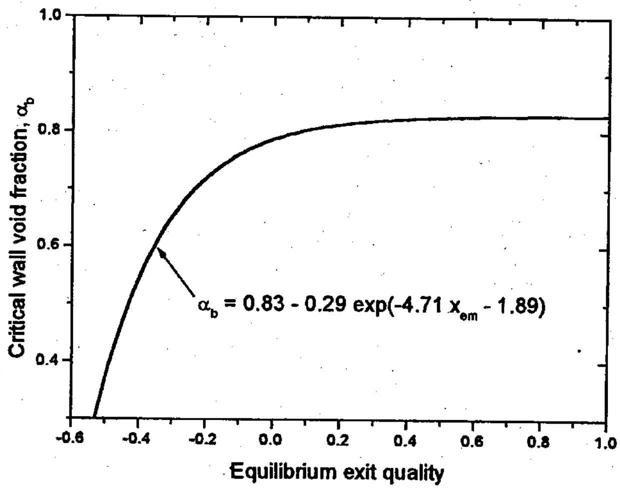

Fig. 2. Correlation of critical wall-void fraction.

The critical wall-void fraction α_b is defined as a volume fraction of steam in the wall bubbly layer at which CHF occurs. The basic problem inherent in the bubble crowding model, including the present model, is that there are insufficient data to accurately specify the α_b. Therefore, in most practical analyses, the underlying task in the bubble crowding model is to determine the α_b. A detailed information of the α_b is quite limited both experimentally and theoretically until now. At this point, we have no better evidence to utilize anything other than the simple expression by experimental data fitting. Although the α_b may be a complex function of heated surface conditions and flow parameters, the relationship between α_b and x_{em} is established, as a first approximation, in collaboration with experimental CHF data by recognizing that the α_b is a function of the exit equilibrium quality. The final choice of α_b incorporated in the present model is shown by the curve in Fig. 2, where it is approximately correlated by the following relation.

$$\alpha_b = 0.83 - 0.29\exp(-4.71x_{em} - 1.89) \tag{20}$$

This equation is valid only when the average bulk void fraction (α_{avg}) at the CHF occurrence is not greater than 0.8, otherwise the flow structure of Fig. 1(a) can not be maintained. The x_{em} instead of the α_{avg} is selected as a parameter in Eq. (20), because it is directly calculated from a heat balance using inlet conditions, while the α_{avg}

is calculated internally using several constitutive models. The physical meaning of this correlation will be discussed in comparisons with experimental CHF data later. It is worth noting that values of 0.82 and 0.75 were utilized by Weisman and Pei and Chang and Lee in their CHF models, respectively. As shown in Fig. 2, the α_b approaches the similar value of the Weisman and Pei's model for high exit quality.

3.3. Constitutive relations

In order to achieve closure of the mass, energy and momentum balance equations, several additional constitutive relations are required. As known well, the accuracy of calculation results by the two-phase flow model is dependent on the utilized constitutive equations. Prediction accuracy of existing theoretical CHF models are dependent on the limited applicability of various constitutive relations employed within them as well as empirical constants.

3.3.1. Bubble detachment point and detached bubble diameter

Reliable predictions of the point of onset of bubble departure (OBD) and the detached bubble diameter are of essential for the prediction of CHF in Eq. (13). The two terms of the OSV and OBD are usually used to mean the same point. Although many predictive models for the OSV in forced convection subcooled boiling have been developed over the past years, the initiation mechanism of significant vapor generation is not well understood yet. The Levy's model (1967) is employed in the present study. According to the critical review by Lee et al. (1992), in which the prediction methods were reviewed and compared with extensive data, the model of Levy is found to be the best one to fit with the whole set of data among the analytical models considered. Levy proposed the required subcooling for the OBD as

$$\Delta h_{\mathrm{sub,\,d}} = q\left[\frac{C_{pf}}{h_{\mathrm{s}}} - \frac{\mathrm{Pr}_f Y_{\mathrm{B}}^+}{G\sqrt{f/8}}\right] \quad \text{for } Y_{\mathrm{b}}^+ \leq 5 \tag{21}$$

$$\Delta h_{\mathrm{sub,\,d}} = q\left[\frac{C_{pf}}{h_{\mathrm{s}}} - \frac{5}{G\sqrt{f/8}}\left[\mathrm{Pr}_f + \ln\left(1 + \frac{\mathrm{Pr}_f Y_{\mathrm{B}}^+}{S} - \mathrm{Pr}_f\right)\right]\right] \quad \text{for } 5 < Y_{\mathrm{B}}^+ \leq 30 \tag{22}$$

$$\Delta h_{\mathrm{sub,\,d}} = q\left[\frac{C_{pf}}{h_{\mathrm{s}}} - \frac{5}{G\sqrt{f/8}}\left[\mathrm{Pr}_f + \ln(1 + 5\mathrm{Pr}_f) + \frac{1}{2}\ln\left(\frac{Y_{\mathrm{B}}^+}{30}\right)\right]\right] \quad \text{for } Y_{\mathrm{B}}^+ > 30 \tag{23}$$

where

$$Y_{\mathrm{B}}^+ = 0.015\frac{(\sigma_f D\rho_f)^{1/2}}{\mu_f} \tag{24}$$

$$f = 0.0055\left[1 + \left(20000\frac{\varepsilon}{D} + \frac{10^6}{\mathrm{Re}_f}\right)^{1/3}\right] \tag{25}$$

and h_{s} is the single phase heat transfer coefficient. The friction factor f adopted in the model of Levy compares well with 4λ of Eq. (27) in the fully turbulent flow regime. The relative roughness parameter of tube ε/D in Eq. (25) was assumed to be 10^{-4} by Levy. Once the subcooling required for the OBD is known using Eqs. (21)–(25), the equilibrium quality at that point is thus calculated by the relation of $x_{\mathrm{d}} = -\Delta h_{\mathrm{sub,\,d}}/h_{\mathrm{fg}}$.

The bubbles grow up on the surface, detach and new bubbles grow again, so that the system is not a steady state. For simplicity, the time-averaged diameter of the growing bubble is considered as the bubble departure diameter. The diameter of wall-attached bubble is assumed to be the same as bubble departure diameter, which is determined from a balance of the fluid forces acting on the bubble and the surface force. Levy (1967) expressed it as

$$D_{\mathrm{b}} = 0.015\sqrt{\frac{8\sigma D\rho_{\mathrm{avg}}}{fG^2}} \tag{26}$$

For some high heat flux situations when the liquid enthalpy at the tube inlet is already above the value of $\Delta h_{\mathrm{sub,\,d}}$, bubbles are assumed to start growing at the tube inlet.

3.3.2. Drag force exerted on the wall-attached bubbles

It is assumed that bubbles attached on the heated surface act as a surface roughness equivalent to bubble size. Chang and Lee (1989) used the friction factor model of Nedderman and

Shearer (1964) with the idea that the bubbly layer might behave like the regular sand roughness used by Nikuradse in his experiment. However, the wall bubbly layer probably acts more like the random roughness found in commercial tubes and piping, because the bubbles are growing and collapsing on the heated wall. Therefore, the turbulent skin friction coefficient λ is calculated using the Colebrook–White equation with a two-phase Reynolds number ($\mathrm{Re}_{2\phi} = GD/\mu_{2\phi}$) to account for the variation of the fluid viscosity near the heated wall, defined as

$$\frac{1}{\sqrt{\lambda}} = 3.48 - 4\log\left(\frac{2D_b}{D} + \frac{9.35}{\mathrm{Re}_{2\phi}\sqrt{\lambda}}\right) \tag{27}$$

Chen (1971) and Taylor and Hodge (1992) reported that the discrete element method (DEM) incorporates more of the basic physics of the interaction between the fluid and the roughness elements than the equivalent sandgrain roughness approach proposed first by Schlichting. In the DEM approach, the flow resistance of a rough surface is divided into two components: one due to the form drag on the roughness element and the other due to the viscous shear on the smooth surface area between the roughness elements. Therefore, the skin friction coefficient λ at the wall is written as

$$\lambda = \frac{\beta\tau_{w,v} + 0.5C_d\rho_{eff}\bar{U}_{eff}^2}{0.5\rho_c\bar{U}_c^2} \tag{28}$$

The drag force term is expressed in the integral form in the original equation (Taylor et al., 1985). However for simplicity in this calculation, it is assumed to be given as in Eq. (28), where C_d is the frictional drag coefficient. The effective variables of velocity and density to bubbles are approximated as mean core values. At near CHF condition, β approaches zero because the heated surface is composed of bubbles compacted in a most dense array. Therefore the drag coefficient becomes to be equal to the skin friction coefficient, i.e., $C_d \cong \lambda$, which is the same approach as done in the subcooled boiling model of Staub (1968). The drag force on a single bubble F_d can be written as

$$F_d \approx \frac{\lambda\rho_c\bar{U}_c^2}{2}\left(\frac{\pi D_b^2}{4}\right) \tag{29}$$

3.3.3. Average velocities of the wall bubbly layer and core region

The average velocity of the wall bubbly layer $\bar{U}_b$ is determined by taking it as half the velocity of the core at the outer edge of the layer, because the velocity profile in the wall bubbly layer is assumed to be linear due to its very thin thickness.

$$\bar{U}_b = 0.5U_c(\text{at } y = D_b) \tag{30}$$

where y is a distance from the wall in the radial direction. The universal logarithmic velocity profile for a single-phase turbulent flow proposed by Karman is assumed to be valid in the turbulent core region.

$$U^+ = Y^+ \quad \text{for } Y^+ < 5 \tag{31}$$

$$U^+ = 5\ln Y^+ - 3.05 \quad \text{for } 5 \le Y^+ < 30 \tag{32}$$

$$U^+ = 2.5\ln Y^+ + 5.5 \quad \text{for } Y^+ \ge 30 \tag{33}$$

where

$$U^+ = U_c\sqrt{\frac{\rho_c}{\tau_w}} \tag{34}$$

$$Y^+ = \frac{D_b\sqrt{\tau_w\rho_c}}{\mu_{2\phi}} \tag{35}$$

$$\tau_w = \frac{\lambda G^2}{2\rho_c\eta_c^2} \tag{36}$$

Once the average velocity of the bubbly layer is determined from Eq. (30), the average velocity of core can be calculated as

$$\bar{U}_c = \frac{G - \bar{U}_b\rho_b(1-\eta_c)}{\rho_c\eta_c} \tag{37}$$

from the constant mass flow rate condition.

$$G = G_b(1-\eta_c) + G_c\eta_c \tag{38}$$

3.3.4. Nonequilibrium flow quality and void fraction

The flow quality and void fraction in the subcooled flow boiling can be evaluated by both the profile-fit and mechanistic approaches (Lahey and Moody, 1993). Because little difference in the CHF predictions appears when both approaches are applied to the present CHF model, the simple profile-fit model of Saha and Zuber (1974) is employed. The relationship between the true flow quality x_{avg} and the thermodynamic equilibrium qualities is expressed by

$$x_{avg} = \frac{x_{em} - x_d \exp(x_{em}/x_d - 1)}{1 - x_d \exp(x_{em}/x_d - 1)} \quad (39)$$

where x_d and x_{em} are thermodynamic equilibrium qualities determined at the OSV point and tube exit, respectively.

Once the flow quality at the tube exit is determined using Eq. (39), the average bulk void fraction α_{avg} is calculated using the Zuber and Findlay model, modified by Dix (1971), that is reasonably accurate for many practical applications. Zuber and Findlay (1965) first proposed the void fraction based on the drift flux model as follows.

$$\alpha_{avg} = \frac{x_{avg}}{\left\{C_0\left[x_{avg} + \frac{\rho_g}{\rho_l}(1 - x_{avg})\right] + \frac{\rho_g V_{gj}}{G}\right\}} \quad (40)$$

In an attempt to relate C_0 and V_{gj} to the void fraction or steam quality, a functional form for the distribution parameter C_0 was suggested by Dix. The Dix model is based on the following volume quality defined as

$$\gamma = \frac{x_{avg}}{\left[x_{avg} + \frac{\rho_g}{\rho_l}(1 - x_{avg})\right]} \quad (41)$$

with γ known, the drift flux parameters C_0 and V_{gj} are given by

$$b = \left(\frac{\rho_g}{\rho_l}\right)^{0.1} \quad (42)$$

$$C_0 = \gamma\left[1 + \left(\frac{1}{\gamma} - 1\right)^b\right] \quad (43)$$

$$V_{gl} = 2.9\left[\frac{(\rho_l - \rho_g)\sigma_l g}{\rho_l^2}\right]^{1/4} \quad (44)$$

4. Verification of the CHF model

4.1. Method of evaluating prediction performance

The procedure of CHF prediction is as follows. For the given geometry of a tube, the conditions of pressure, inlet flow rate and inlet subcooling are fixed, then an assumed heat flux to the tube wall is varied until the assumed and calculated heat fluxes converge into one value. Since the model presented in this paper is complicated, the CHF calculation procedure is presented in Appendix B. The critical heat fluxes predicted by the present model are compared with the experimental CHF data. The results are quantitatively evaluated by the CHFR, defined as

$$\mathrm{CHFR} = \frac{\mathrm{CHF_p}}{\mathrm{CHF_m}} \quad (45)$$

where subscripts p and m mean predicted and measured values, respectively. The three statistical parameters of μ (average value), σ (sample standard deviation) and RMS (root-mean-square error) are defined as

$$\mu = \frac{1}{N}\sum_{i=1}^{N}(\mathrm{CHFR}_i) \quad (46)$$

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}[\mu - \mathrm{CHFR}_i]^2} \quad (47)$$

$$\mathrm{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}[1 - \mathrm{CHFR}_i]^2} \quad (48)$$

where CHFR_i is the CHFR of the i'th data and N stands for the total number of data points predicted successfully by the present model.

4.2. Comparisons with experimental data

To assess the predictive capabilities and the limitations of the proposed model, a total of 5009 water CHF data points, covering a wide range of different parameters, was selected from the KAIST CHF database (Chang et al., 1996). The database comprises of about 15 000 data for water flow in uniformly heated vertical round tubes compiled from various sources. The range of parameters for each data source, considered in this study, is presented in Table 1 (Thompson and Macbeth, 1964; Tong, 1964; Mayinger, 1967; Becker et al., 1971; Zenkevich, 1974). These data points, satisfying the condition of $\alpha_{avg} \leq 0.8$ at the tube exit, are selected by using Eqs. (21)–(23), (39) and (40) before beginning the CHF calculation. The parameter ranges of the selected data

Table 1
Experimental conditions of CHF data used for fitting the correlation of critical wall-void fraction

Reference	Parameter							
	N	D (m)	L (m)	P (MPa)	G (kg m^{-2} s^{-1})	$\Delta h_{sub, in}$ (kJ kg^{-1})	x_{em}	q_{CHF} (kW m^{-2})
Thompson and Macbeth (1964)	1122	0.0011∼0.038	0.035∼1.97	2.14∼18.96	496∼7499	2.0∼1659	−0.49∼0.41	760∼14 800
Becker et al. (1971)	804	0.004∼0.02	0.4∼4.97	3.04∼20.0	452∼7438	60∼1372	−0.31∼0.60	135∼7480
Zenkevich (1974)	2925	0.005∼0.011	1.0∼6.0	5.89∼19.62	497∼6694	5∼1621	−0.49∼0.62	137∼7290
Tong (1964)	123	0.006∼0.013	0.38∼3.66	5.17∼13.79	678∼7499	27∼1060	0.00∼0.38	760∼5900
Mayinger (1967)	35	0.007	0.56∼0.98	7.0∼10.24	2255∼3734	0∼278	0.098∼0.21	924∼2860
Total	5009	0.0011∼0.038	0.035∼6.0	2.14∼20.0	452∼7499	0∼1659	−0.49∼0.62	135∼14 800

points are diameters from 1 to 37.5 mm, lengths from 0.035 to 6 m, mass fluxes from 450 to 7500 kg m^{-2} s^{-1}, pressures from 2 to 20 MPa, inlet subcooling from 0 to 1660 kJ kg^{-1} and critical heat fluxes from 135 to 14 800 kW m^{-2}, which covers the operating ranges of typical light water reactor (LWR).

A set of α_b as a function of x_{em}, resulting in an average CHFR of 1.0 while minimizing RMS error, were investigated by varying its value against the data points of Table 1. As a result, the correlation of Eq. (20) was obtained as shown in Fig. 2. Fig. 3 shows the predicted CHF as a function the α_b, where the corresponding x_{em} is presented in the top of x-axis. The α_b decreases as the CHF increases even though some scattered data exist. This trend is consistent with the experimental observation by Styrikovitch et al. (1970) that the wall void fraction at CHF dropped significantly at very high subcooling and high heat fluxes. It is also consistent with the general understanding that CHF could occur in low wall-voidage in the case of very high heat flux (Collier and Thome, 1994).

Fig. 4 shows the visual comparison of the predicted and measured CHF using the above data set. Most of the experiment data (about 93%) is successfully predicted within $\pm 20\%$ error band. The standard deviation σ and RMS error of CHFR are 11.12 and 11.14%, respectively. It is interesting that most of data is well predicted within $\pm 20\%$ when CHF is less than 1 MW m^{-2} as shown in Fig. 4. Some data points of Thompson and Macbeth and Tong, however, deviate from the error range, which may be considered as typical uncertainties associated with experiments. It should be noted that even though Hall and Mudawar (1997) found several

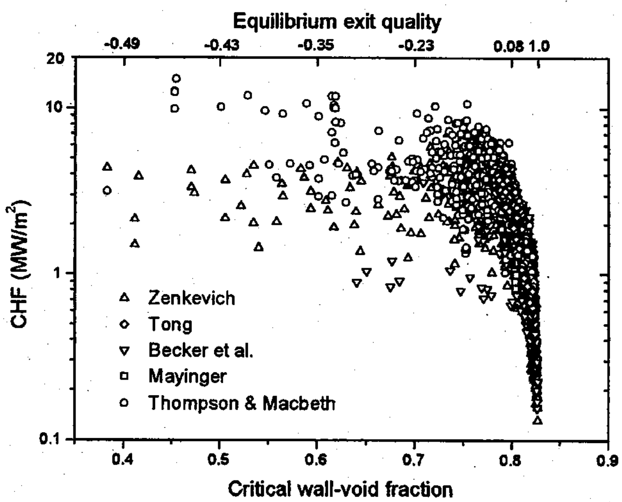

Fig. 3. CHF vs. critical wall-void fraction.

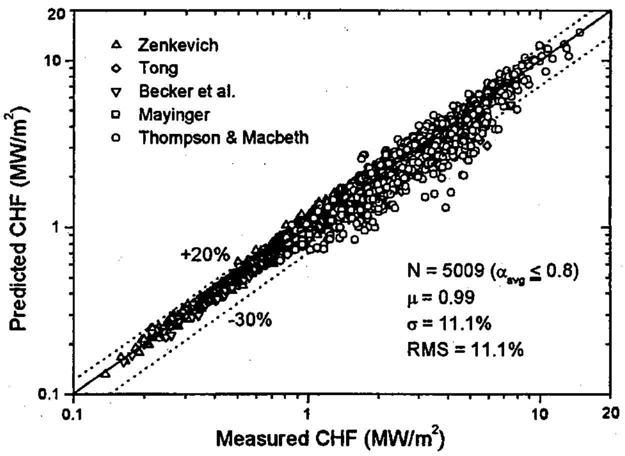

Fig. 4. Predicted vs. experimental CHF.

problems in the database of Thompson and Macbeth, no examination was conducted to re-evaluate all their data in this study. Most of the deviating data from $\pm 20\%$ error band belongs to their database. If we restrict the application range of exit void fraction less than 0.7, about 97% of the total 3498 data is predicted within $\pm 20\%$, with $\mu = 1.02$, $\sigma = 8.91\%$ and $\mathrm{RMS} = 9.2\%$, while 99% data is predicted within $\pm 30\%$ error band.

For the subcooled flow boiling with small inside diameter and high liquid velocity, generally, the pressure drop along the axis of flow becomes high. For these conditions, the assumption of neglecting the axial pressure drop in mass and energy balance equations is considered to be rough approximation. In the KAIST database, since most of the CHF data points with small diameter tube, satisfying $D \leq 3$ mm, does not have very high velocity, the proposed model predicted CHF with acceptable accuracy. It will be discussed in detail in Section 4.4.

Table 2 shows the statistical parameters of CHFR predicted by the present model. The prediction accuracy is very good for $0.1 \leq \alpha_{avg} < 0.7$, while it deteriorates for highly subcooling ($\alpha_{avg} < 0.1$) and high quality regions ($\alpha_{avg} \geq 0.7$). It should be noted that the classification of data points in Table 2 are conducted by using Eqs. (21)–(23), (39) and (40) based on experimental condition, because the average bulk void fractions at the tube exit of the experiment and predicted result may be different.

The dependence of the prediction accuracy on major parameters is presented in Figs. 5–10, where the CHFR is plotted versus pressure P, mass flux G, inlet subcooling $\Delta h_{sub,\,in}$, average bulk void fraction at tube exit α_{avg}, tube diameter D and ratio of tube length-to-diameter L/D, respectively. A comparison of theoretical predictions by the present CHF model with experimental data does not seem to exhibit significant systematic deviations which could be attributed to a certain system parameter, such as thermal hydraulic conditions (P, G, $\Delta h_{sub,\,in}$, α_{avg}) and geometric parameters (D and L/D). However, the predictions at the low pressure ($P < 7$ MPa) and low mass flux ($G < 1500$ kg m^{-2} s^{-1}) regions in the KAIST database are generally underestimated as shown in Fig. 5 and Fig. 6, respectively.

Table 2
Statistical results of CHFR depending on the average bulk void fraction at the tube exit

Range of α_{avg}	N	μ	σ (%)	RMS (%)	$\pm 20\%$ error (%)	$\pm 30\%$ error (%)
$0.1 > \alpha_{avg} \geq 0$	26	1.077	9.97	12.46	84.6	96.2
$0.2 > \alpha_{avg} \geq 0.1$	73	1.039	8.17	8.99	94.5	100
$0.3 > \alpha_{avg} \geq 0.2$	133	1.042	9.35	10.22	95.5	98.5
$0.4 > \alpha_{avg} \geq 0.3$	318	1.021	7.04	7.35	97.5	100
$0.5 > \alpha_{avg} \geq 0.4$	626	1.041	8.74	9.64	95.4	98.7
$0.6 > \alpha_{avg} \geq 0.5$	901	1.036	8.28	9.03	96.6	99.9
$0.7 > \alpha_{avg} \geq 0.6$	1288	1.007	9.28	9.30	97.2	98.8
$0.8 \geq \alpha_{avg} \geq 0.7$	1644	0.928	12.31	14.26	84.8	94.0

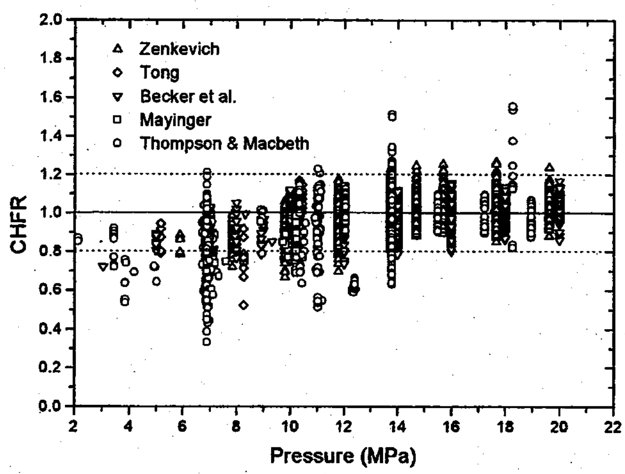

Fig. 5. CHFR vs. pressure.

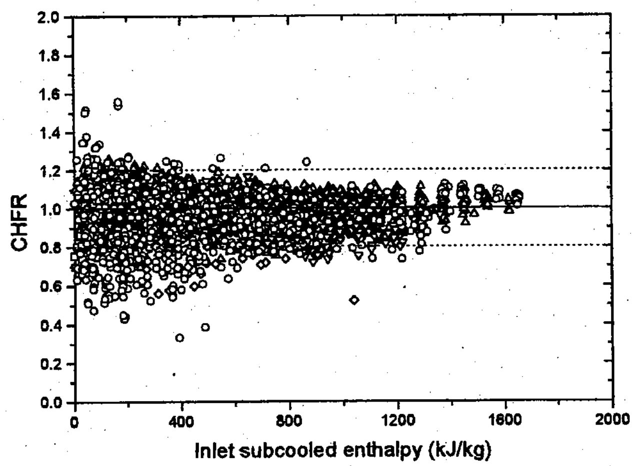

Fig. 7. CHFR vs. inlet subcooling enthalpy.

It was observed in the preliminary analyses that as the mass flux decreases below 450 kg m^{-2} s^{-1} the prediction accuracy significantly deteriorates for some cases. This indicates the inadequacy of the relevant constitutive relations included in the present model for low mass flow rate conditions.

For comparing with the present model, the predictions using the CHF look-up table of Groeneveld et al. (1996) were performed based on the so-called heat balance method (HBM). Because the look-up table has its applicable ranges, 5006 out of 5009 data points were used for comparison. For tubes of diameter other than referenced 8 mm, the diameter correction rule suggested by the table authors was used. The average value of the CHFR is 1.02. The standard deviation and RMS error are 7.93 and 8.11%, respectively. Fig. 11 shows the comparisons of 5006 data points with the predictions by the look-up table. The percentage of data points calculated with a given error band ($\pm$%) for both the look-up table and the present model is presented in Fig. 12. As shown in the figure, the prediction accuracy of the look-up table is better than the present model for the entire range. However, for subcooled conditions when the equilibrium quality at the tube exit is less than zero, the prediction accuracy of the present model is better than the

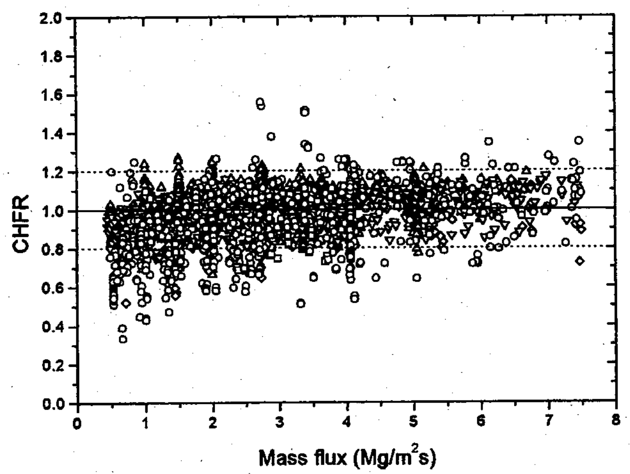

Fig. 6. CHFR vs. mass flux.

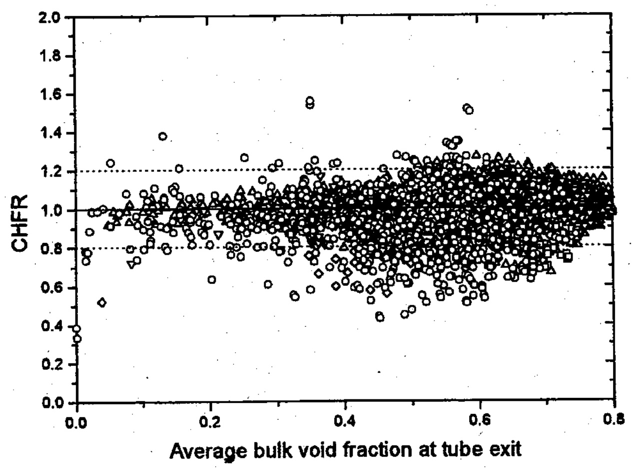

Fig. 8. CHFR vs. average bulk void fraction at the tube exit.

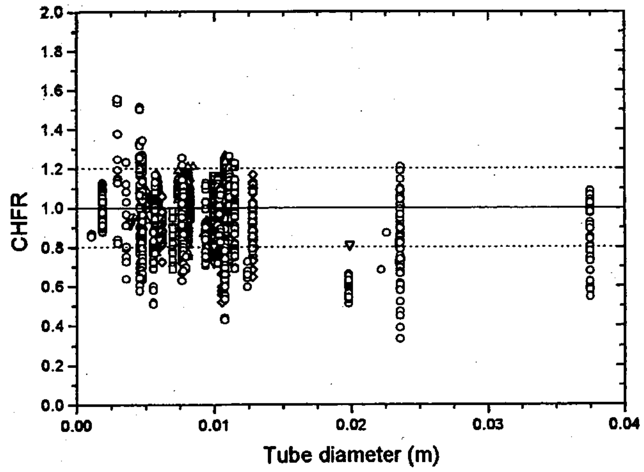

Fig. 9. CHFR vs. tube diameter.

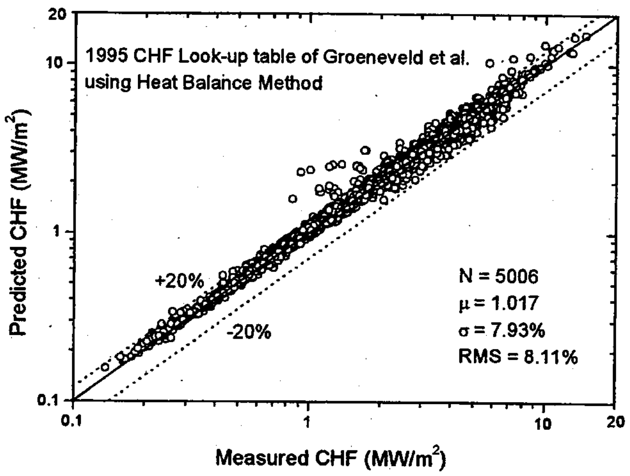

Fig. 11. CHF predictions by the look-up table of Groeneveld et al.

CHF look-up table. About 96% of the total 902 data is predicted within $\pm 20\%$ error band, with $\mu = 0.995$, $\sigma = 8.75\%$, $\mathrm{RMS} = 8.76\%$, respectively. While the Lookup table of Groeneveld et al. predicts 93% of the 899 data within $\pm 20\%$, with $\mu = 1.02\%$, $\sigma = 10.97\%$, $\mathrm{RMS} = 11.12\%$, respectively.

4.3. *Parametric trends of system variables*

Fig. 13 shows that the parametric trends of CHF predicted by the present model are in good agreement with the data of Weatherhead (1963) (obtained from the book of Collier and Thome, 1994). Also shown in Fig. 13 is the influence of inlet subcooling on CHF in the relatively high pressure region (13.8 MPa). The CHF increases almost linearly with inlet subcooling but its effect decreases with decreasing mass flux. The trends agree with general understanding regarding the CHF characteristics. The effect of liquid subcooling on CHF is prominent because the subcooled liquid flow condenses steam bubbles and suppresses their coalescence on the wall.

Fig. 14 shows the effect of mass flux on CHF with the pressure as another independent variable, where CHF increases with increasing mass flux. The effect of mass flux depends on the system

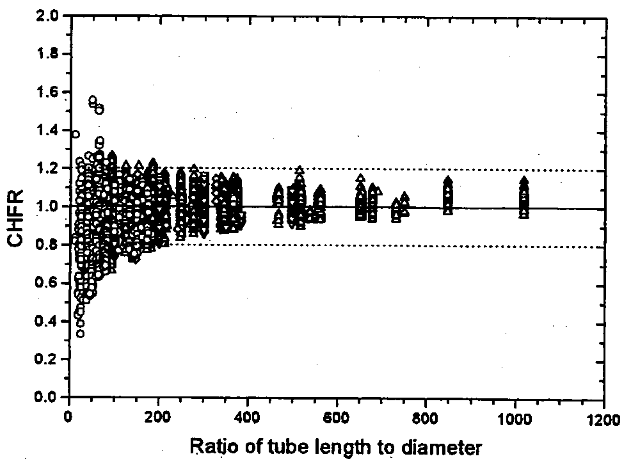

Fig. 10. CHFR vs. ratio of tube length to diameter.

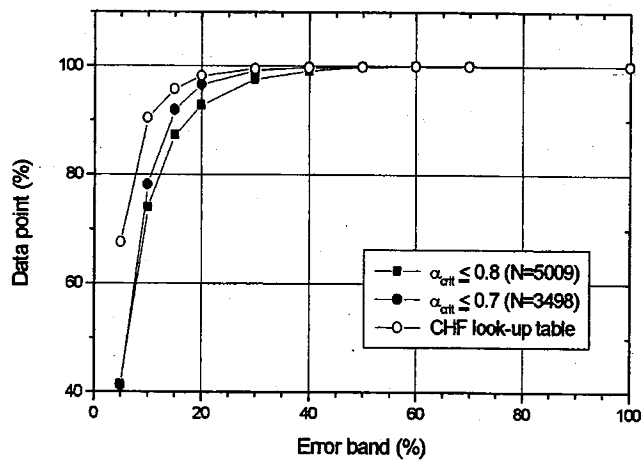

Fig. 12. Percentage of data points predicted within error band for the present model and the 1995 CHF look-up table of Groeneveld et al.

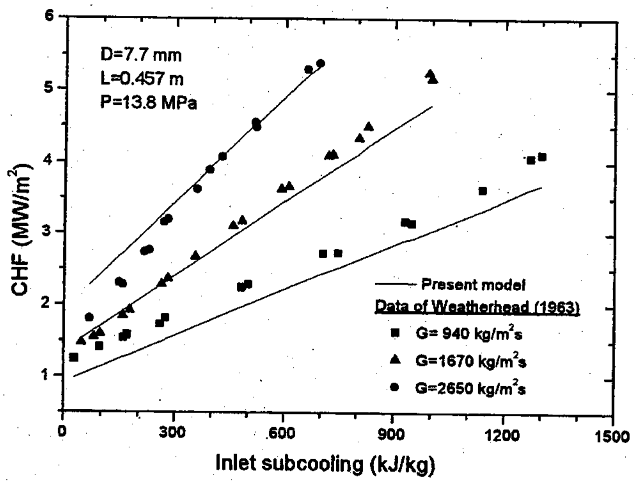

Fig. 13. Comparison of predicted CHF with experimental data of Weatherhead (1963).

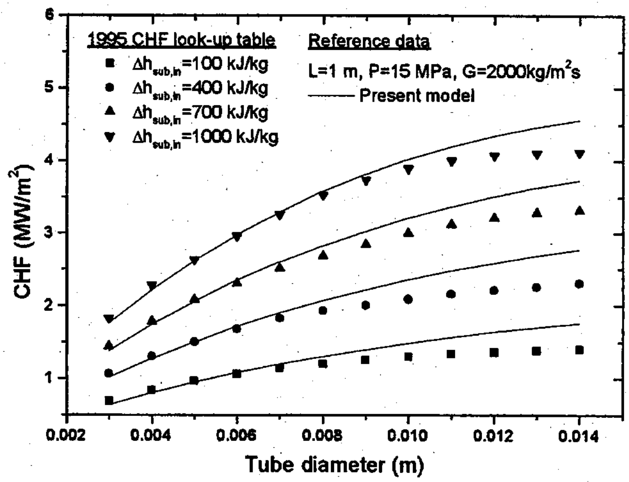

Fig. 15. Prediction of tube diameter effect on CHF for fixed inlet condition compared with the 1995 CHF look-up table.

pressure such that it is stronger at lower pressure regions. The experimental data in Fig. 14 are directly taken from the CHF look-up table of Groeneveld et al. (1996) for an 8 mm diameter tube based on reference exit quality, $x_{em} = -0.2$.

The effect of tube diameter on CHF is predicted by the present model, together with the interrelation between tube diameter and inlet subcooling as shown in Fig. 15. The CHF increases with increasing tube diameter at fixed inlet conditions. The effect increases with the increase of the inlet liquid subcooling. Because the CHF look-up table presents CHF values for 8 mm tubes, a diameter correction factor for CHF is used to extend the applications to other values of tube diameter. Groeneveld et al. (1996) provided the following relationship between the CHF and the tube diameter based on fixed local conditions.

$$\frac{\mathrm{CHF}_D}{\mathrm{CHF}_{D=8\,\mathrm{mm}}} = \left(\frac{D}{8}\right)^{-1/2} \tag{49}$$

where CHF_D is the CHF for a diameter of interest, $\mathrm{CHF}_{D=8\,\mathrm{mm}}$ is the CHF for an 8-mm tube and D is the tube diameter in mm. However, it has been recognized that the diameter effect on CHF is very complex. For small tube diameters, the prediction by the present model agrees well with the CHF look-up table, however, the difference between two predictions increases as the tube diameter increases.

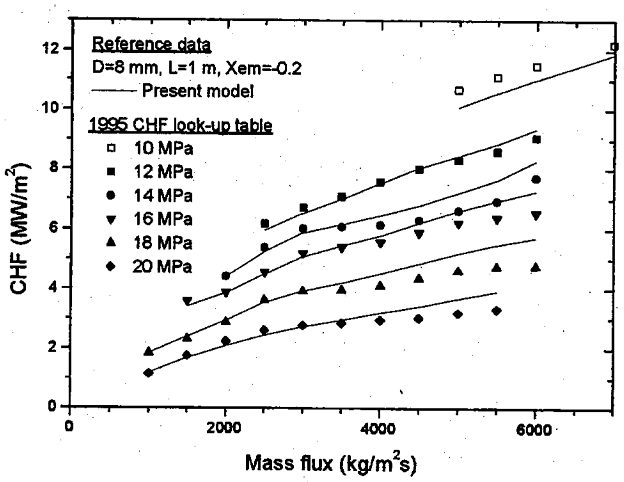

Fig. 14. Prediction of mass flux effect on CHF compared with the 1995 CHF look-up table.

It is a general understanding that CHF is a decreasing function of tube length for fixed inlet conditions. The present model follows well the general trends of experimental data as shown in Fig. 16. It is noted that data points shown in the figure are only for the cases where the 1995 CHF look-up table data are available with a heat balance equation. The CHF decreases more rapidly for short tubs and the length effect has small contribution to CHF for longer tubes. The ratio of the heated length to the tube diameter of the channel (L/D) is the characteristic non-dimen-

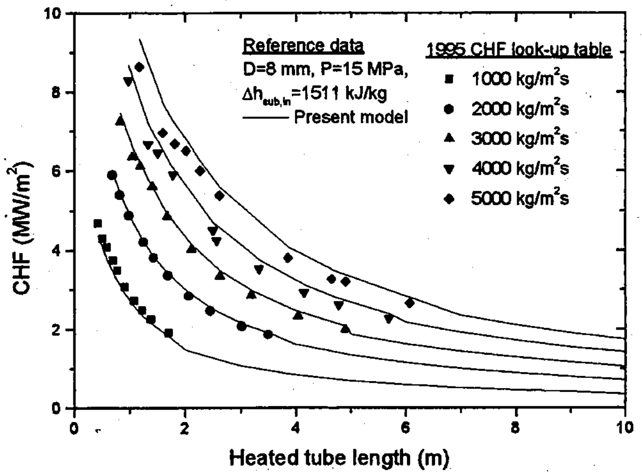

Fig. 16. Prediction of heated length effect on CHF for fixed inlet condition compared with the 1995 CHF look-up table.

sional length for the CHF investigation. However, it still needs to be established (Celata, 1998). The threshold value of L/D beyond which it does not affect CHF does not appear for fixed inlet conditions, in which CHF is always a decreasing function of L/D as shown in Fig. 16 (Hewitt, 1982; Moon et al., 1996).

The parametric trends shown in Figs. 13–16 demonstrate that the present model provides a reliable accuracy in predicting independent CHF variations with respect to mass flux, pressure, inlet subcooling, tube diameter and tube length.

4.4. Validation of the CHF prediction procedure

For the validation of the proposed CHF prediction method, we assessed the model against other CHF databases that not used for fitting the critical wall-void fraction of Eq. (20). The parameter ranges of new databases for water and non-aqueous liquids are presented in Table 3 with references. Three water databases of Kutateladze (1969), Chen et al. (1997) and ENEA (Celata et al., 1994a,b) were utilized. The operating ranges of Kutateladze are covered by KAIST database, however, Chen et al.'s data have low pressure condition beyond the valid range of Eq. (20). The previous two databases contain water CHF data for intermediate heat flux from 1 to 20 MW m^{-2}. As shown in Fig. 17, the proposed model well predicts CHF against two databases with

Table 3
Experimental conditions of CHF data selected to validate the present CHF prediction method

Reference	Parameter							
	N	D (m)	L (m)	P (MPa)	G (kg m^{-2} s^{-1})	$\Delta h_{\text{sub, in}}$ (kJ kg^{-1})	x_{em}	q_{CHF} (kW m^{-2})
Kutateladze (1969)	150	0.008 ~ 0.0112	0.4 ~ 3.66	9.07 ~ 20.26	1383 ~ 9494	64.2 ~ 1019	−0.014 ~ 0.165	978 ~ 5442
Chen et al. (1997)	109	0.01 ~ 0.016	0.295 ~ 0.4	0.146 ~ 1.66	1427 ~ 13397	228 ~ 701	−0.236 ~ −0.058	4170 ~ 14 560
Celata et al. (1994a,b)	1887	0.0003 ~ 0.0254	0.0025 ~ 0.61	0.1 ~ 8.4	1000 ~ 90000	25 ~ 255*	−0.516 ~ 0.088	3333 ~ 228 000
Stevens and Miles (1980)**	27	0.0153 ~ 0.0213	2.85 ~ 3.94	0.98 ~ 1.09	2402 ~ 3868	7.9 ~ 35.1	−0.012 ~ 0.108	108 ~ 193
Merilo and Ahmad (1979)**	63	0.0053, 0.0126	1.03 ~ 3.66	1.04 ~ 1.53	2698 ~ 8163	3.2 ~ 40.2	0.013 ~ 0.11	124 ~ 417
Coffield et al. (1969)***	40	0.01021	0.762	0.91 ~ 2.19	1288 ~ 5601	11.6 ~ 103.3	−0.55 ~ 0.025	136 ~ 557

* Inlet subcooling in temperature (K).
** CHF database for Freon-12.
*** CHF database for Freon-113.

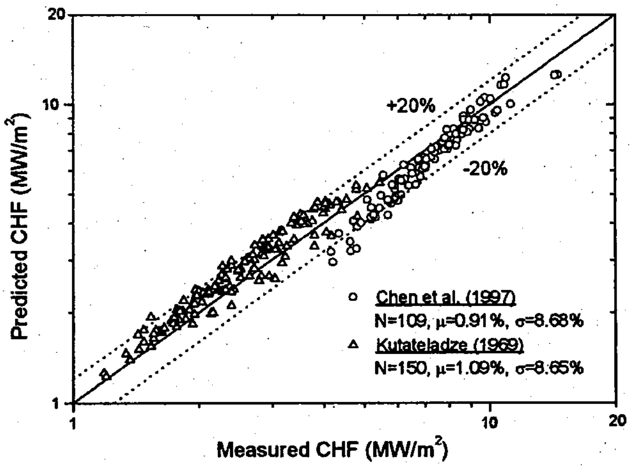

Fig. 17. Comparison of predicted CHF with the experimental data of Kutateladze (1969) and Chen et al. (1997).

similar statistical results compared to the KAIST database, even though the operating pressure is out of the applicable range. The model was proved out again that it has a tendency to underpredict within 20% error band for low pressure ranges

A subcooled flow boiling has been widely investigated for the thermal hydraulic design of LWRs, where the magnitude of heat fluxes is generally an order of MW m^{-2}. However, for the application to fusion reactor components, very high heat removal rate, being an order of magnitude higher than LWRs, is required. The thermal hydraulic conditions required are such as high subcooling, high mass flux (greater than 10 Mg m^{-2} s^{-1}), small tube diameter, low-intermediate pressure (0.1–5 MPa) and very high heat flux (up to 80 MW m^{-2}). The liquid sublayer dryout model, proposed by Celata et al. (1994a), is known to be capable of predicting the CHF of highly subcooled water flow boiling in a low-intermediate pressure region.

The proposed model was tested on the ENEA database satisfying the above extreme conditions. The ENEA database comprises of 1887 data points compiled from many different references. Nariai and Inasaka (1992) reported the abnormality of the subcooled flow boiling in the narrow tubes with high velocity (less than 3 mm diameter with higher than 70 000 kg m^{-2} velocity). According to their experiment, the pressure drop and actual void fraction for the narrow tubes with high liquid velocity become considerably different from those estimated by the existing correlations or models. For the reason mentioned above, we have divided the ENEA CHF data into two groups. In the first group of $D \leq 3$ mm, 984 data points out of 1887 were predicted with $\mu = 0.84$, $\sigma = 23.3\%$. Total 210 data points in the first group have less than 1 mm inside diameter. While for the other group of $D > 3$ mm, 903 data points were well predicted with $\mu = 0.97$, $\sigma = 14.45\%$. The comparison of the predicted CHF by the proposed model and experimental data is presented in Fig. 18.

The proposed prediction method was compared with the non-aqueous liquids experimental data obtained for Freon-12 by Stevens and Miles (1980) and Merilo and Ahmad (1979), for Freon-113 by Coffield et al. (1969), respectively. The selected data points satisfy that the exit average bulk void fraction is less than 0.8. Fig. 19 shows a relatively good agreement between the predictions and experimental data. The operating ranges of experimental parameter are presented in Table 3.

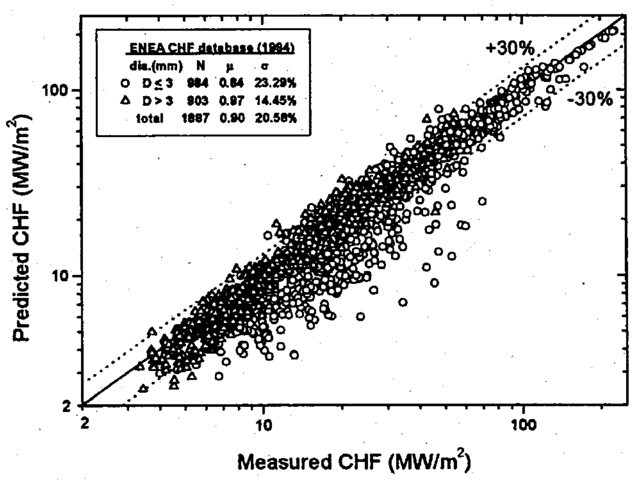

Fig. 18. Comparison of predicted CHF with the expeninental data of ENEA (Celata et al., 1994a,b) for highly subcooled flow boiling.

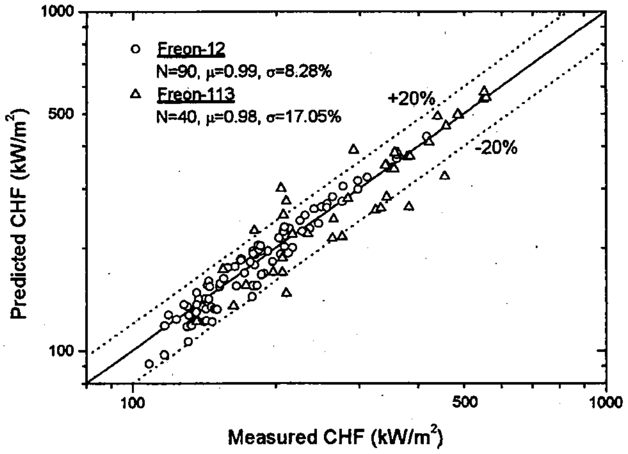

Fig. 19. Comparison of predicted CHF with the experunental data of Freon-12 and 113 for $\alpha_{avg} < 0.8$.

The assessment of the proposed model against new CHF databases, including data beyond the applicable ranges and non-aqueous liquids, shows a satisfactory agreement with similar statistical results compared to the KAIST database. Therefore, the ability of the proposed model to correctly predict new CHF databases increases the confidence of this model.

5. Discussions

It is suggested here that the lateral coalescence of bubbles on the wall take place if the wall-void fraction reaches a certain critical value as heat flux approaches CHF. The effect of coalescence will eventually result in formation of large vapor clots. When a large merged bubble is created in the wall bubbly layer, the flow structure near the wall becomes similar to that of the liquid sublayer dryout model. In the sublayer dryout model, the stationary tiny vapor jets anchored to active sites on the heated wall provide vapor to the slug bubble. The present model focuses on the conditions just before bubbles in the wall bubbly layer coalesce into a large vapor clot. While the sublayer dryout model focuses on the conditions just after formation of a large vapor clot, which may happen to occur following the former process as pointed out by Tong and Weisman (1996). Because the CHF can be approached from opposite directions on the boiling curve, both approaches may be satisfactory as a basis for CHF prediction.

We have phenomenologically described the CHF mechanism, however, basic mechanisms forming the constitutive relations adopted are so complicated that many uncertainties and inaccuracies could be included in the CHF model. Some of these constitutive models could be incorrect, which would naturally bias the choice of the arbitrary critical wall-void fraction α_b. It could compensate errors due to wrong assumptions made in the CHF model. However, considering the current state-of-the-art in the experiments and models of subcooled and low quality flow boiling followed by the CHF, it is not possible to synthesize a mechanistic model with complete generality in the present time. Assuming the validity of the functional forms of the constitutive relations employed in the proposed model, we may optimize the choice of the $_b$ in collaboration with available experimental CHF data as shown in Fig. 2. We have ignored the microscopic phenomena of bubble dynamics on the heated wall and the statistical variations in bubble size and population density as well as various flow parameters. The basis of the critical wall-void fraction should be derived from the theoretical investigation with the aid of experimental observation in the future.

The CHF formula of Eq. (13) allows the prediction of the CHF as a function of fluid transport properties and geometry parameters. The present model takes into account the convective shear effect due to the drag force on the wall-attached bubbles, which represents the characteristic parameter for turbulent interchange between the wall bubbly layer and core. Even though the constitutive relations employed in the model are based on the simplifying assumptions, it is interesting to see that the CHF model works well in comparison with experimental data, including non-aqueous liquids and water data beyond the valid ranges of Eq. (20). This obviously results from the fact that the main contribution to the subcooled and low quality flow boiling CHF normally stems from limiting thermal transportation by the transverse mass interchange at the outer edge of the wall bubbly layer. The effect is well described by the momentum balance equation in conjunction with the critical wall-void fraction in the present model.

6. Conclusion

A mechanistic model based on the concept of lateral bubbles coalescence in the wall bubbly layer has been developed to predict the CHF for subcooled and low quality flow boiling in a uniformly heated vertical tube. Adopting the Chang and Lee (1989) approach utilizing mass, energy and momentum balance equations, an improved CHF formula has been derived with more solid theoretical bases. Results are summarized as follows:

1. Comparison of the current model predictions to the experimental CHF data shows a good agreement over a wide range of parameters for both light water and fusion reactors operating conditions. An empirical correlation for the critical wall-void fraction was obtained by data fitting against the KAIST CHF database. With use of it, a relatively good prediction has been performed within ±20% error band. The overall mean ratio of the predicted to measured CHF values is 0.99 with a standard deviation of 11.12% and a RMS error of 11.14% against about 5000 water CHF data points.
2. The proposed model has been validated against different CHF databases that not used for fitting of the critical wall-void fraction. For the databases comprising of water CHF for typical fusion reactor and non-aqueous liquids, comparison results show a relatively good prediction accuracy.
3. The limiting transverse interchange of mass flux crossing the interface of the wall bubbly layer and core is represented well by taking account of the convective shear effect due to the drag force on the wall-attached bubbles. It could be said that the dominant mechanism controlling the CHF in the subcooled and low quality flow boiling is properly represented in the present model.

It is suggested that refinement of this model should be pursued in the future to improve the critical wall-void fraction. Furthermore, an improvement in the current model can readily be made if better constitutive laws are used for the complex boiling phenomena.

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Appendix A. Nomenclature

A	cross-section area (m^2)
b	density ratio defined by Eq. (42)
C_d	local drag coefficient
C_{pf}	specific heat (J kg^{-1} K^{-1})
C_o	distribution parameter
D	tube diameter (m)
D_b	detached bubble diameter (m)
f	skin friction factor
F_d	drag force on a single bubble (N)
F_q	fraction of heat flux producing vapor
G	mass flux (kg m^{-2} s^{-1}), $G_b = P_bU_b$, $G_c = P_cU_c$
G^*	limited transverse mixing mass flux (kg m^{-2} s^{-1})
g	acceleration due to gravity (ms^{-2})
h	enthalpy (kJ kg^{-1})
h	degree of subcooling (kJ/kg)
h_s	single phase heat transfer coefficient (W m^{-2} K^{-1}) ($=0.023Re^{0.8}$ $Pr^{0.4}K/D$)
h_{fg}	latent enthalpy of vaporization (kJ kg^{-1})
K	thermal conductivity (W mK^{-1})
N_{bub}	number of bubble
P	pressure (MPa)
Pr	Prandtl number
q	heat flux (kW m^{-2})
Re	Reynolds number
RMS	root-mean-square error
r_o	radius of tube (m)

U local velocity (m s^{-1})
U^+ dimensionless velocity
$\bar{U}$ mean velocity (m s^{-1})
V_{gj} drift velocity (m s^{-1})
x quality
x_{em} thermal equilibrium quality at tube exit
y distance in radial direction (m)
Y^+ dimensionless distance
Y_B^+ dimensionless distance to tip of vapor bubble
z axial location (m)

Greek letters
α void fraction
β blockage factor
ε roughness size (m)
γ volume quality defined by Eq. (41)
η_c fraction of cross-section occupied by core
λ skin friction coefficient ($=0.25\,f$)
μ viscosity (Ns m^{-2}) or average mean value
ρ density (kg m^{-3})
σ surface tension (N m^{-1}) or sample standard deviation
τ shear stress (N m^{-2})
τ_w apparent wall shear stress (N m^{-2})
$\tau_{w,v}$ viscous shear stress on wall (N m^{-2})
ξ perimeter (m)
Φ_{acc} acceleration term

Subscripts
avg average
b bubbly layer
bc from bubbly layer to core
c core
cb from core to bubbly layer
CHF CHF location
d bubble detachment position
eff effective
evp evaporation
f saturated liquid
g saturated vapor
i interface of bubbly layer and core
in inlet
l liquid phase
m measured
p predicted
sub subcooled
w heated wall
2ϕ two-phase mixture flow

Appendix B. CHF calculation procedure

Required input parameters: D, L, P, G, $\Delta h_{sub,in}$ (or h_{in}).

1. Assume a value for q.
2. Calculate $\Delta h_{sub,d}$ by Eqs. (21)–(25), where all physical properties are calculated at saturated state at the system pressure.
3. Calculate x_d and z_d.

$$x_d = \frac{-\Delta h_{sub,d}}{h_{fg}}, \quad z_d = \frac{GD(h_f - \Delta h_{sub,d} - h_{in})}{4q}$$

if

$$z_d < 0; \quad z_d = 0 \quad \text{and} \quad x_d = \frac{-\Delta h_{sun,in}}{h_{fg}}$$

4. Calculate x_{em}, h_l, ρ_l and σ_l at the tube exit

$$x_{em} = \frac{h_{avg} - h_f}{h_{fg}}, \quad \text{where} \quad h_{avg} = h_{in} + \frac{4qL}{GD}$$

$$h_l = \frac{h_{avg} - h_g x_{avg}}{1 - x_{avg}}$$

$$T_l = T(P,h_l),\ \rho_l = \rho(P,h_l),\ \sigma_l = \sigma(T_l)$$

5. Calculate x_{avg} by Eq. (39) and α_{avg} by Eqs. (40)–(44).
6. Calculate $\mu_{2\phi}$ by Eq. (19), D_b by Eq. (26) using ρ_{avg} and $Re_{2\phi}$ and η_c by Eq. (11).

$$\rho_{avg} = \frac{\alpha_{avg}\rho_g}{x_{avg}}$$

7. Calculate α_b by Eq. (20), α_c by Eq. (14), ρ_b and ρ_c by Eqs. (15) and (16) and h_b and h_c by Eqs. (17) and (18) using x_b and x_c, respectively.

$$x_b = \frac{\alpha_b\rho_g}{\rho_b}, \quad x_c = \frac{\alpha_c\rho_g}{\rho_c}$$

8. Calculate λ by Eq. (27), τ_w by Eq. (36), $\bar{U}_b$ by Eq. (30), $\bar{U}_c$ by Eq. (37) using Eqs. Eqs. (31)–(35) and F_d by Eq. (29).
9. Calculate q_{CHF} by Eq. (13).

If the estimated q in step (1) is close enough to the calculated q_{CHF} in step (9), q is the critical heat flux. Otherwise, readjust q and return to step (1).

References

Beattie, D.R.H., 1980. An evaluation of two bubble detachment models for two-phase flow, ANS-ASME Topical Meeting on Nuclear Reactor Thermo-hydraulics, Saratoga Springs, NY, NUREG/CP-0014.

Beattie, D.R.H., Whalley, P.B., 1982. A simple two phase frictional pressure drop calculation method. Int. J. Multiphase Flow 8, 83–87.

Becker, K.M., Strand, G., et al., 1971. Round tube burnout data for flow of boiling water at pressure between 30 and 200 bar, Report KTH-NEL-14.

Bricard, P., Souyri, A., 1995. Understanding and modeling DNB in forced convective boiling: a critical review, Two-phase flow modeling and experimentation, Edizioni ETS, pp. 843–851.

Celata, G.P., Cumo, M., Mariani, A., Simoncini, M., Zummo, G., 1994a. Rationalization of existing mechanistic models for the prediction of water subcooled flow boiling critical heat flux. Int. J. Heat Mass Transfer 37 (1), 347–360.

Celata, G.P., Cumo, M., Mariani, A., 1994b. Assessment of correlations and models for the prediction of CHF in water subcooled flow boiling. Int. J. Heat Mass Transfer 37, 237–255.

Celata, G.P., 1997. Modeling of critical heat flux in subcooled flow boiling. Convective Flow and Pool Boiling Conference, Kloster Irsee, Germany, May.

Celata G.P., 1998. Critical heat flux in subcooled flow boiling. In: Proceedings of the 11th International Heat Transfer Conference, Kyongju, South Korea, vol. 1, pp. 261–277.

Chang, S.H., Lee, K.W., 1989. A critical heat flux model based on mass, energy and momentum balance for upflow boiling at low qualities. Nucl. Eng. Des. 113, 35–50.

Chang, S.H., Baek, W.P., Moon, S.K., et al., 1996. The KAIST CHF Data Bank (Rev.3), KAIST-NUSCOL-9601.

Chen, C.K, 1971. Characteristics of turbulent flow resistance in pipes roughened with hemispheres, Ph.D thesis, Washington State University.

Chen, Y., Zhou, R., Hao, L., Chen, H., 1997. Critical heat flux with subcooled boiling of water at low pressure. Eight International Topical Meeting on Nuclear Reactor Thermal-Hydraulics, Kyoto, Japan, vol. 2, pp. 958–964.

Coffield, R.D. Jr., Rohrer, W.M. Jr., Tong, L.S., 1969. A subcooled DNB investigation of Freon-113 and its similarity to subcooled water DNB data. Nucl. Eng. Des. 11, 143.

Collier, J.G., Thome, J.R., 1994. Convective Boiling and Condensation, 3. Clarendon Press, Oxford.

Dix, G.E., 1971. Vapor void fractions for forced convection with subcooled boiling at low flow rates, NEDO-10491, General Electric Company.

Groeneveld, D.C., Leung, L.K.H., Kirillov, P.L., et al., 1996. The 1995 look-up table for critical heat flux in tubes. Nucl. Eng. Des. 163, 1–23.

Gunther, F.C., 1951. Photographic study of surface-boiling heat transfer to water with forced convection. Trans. ASME 73, 115–121.

Hall, D., Mudawar, I., 1997. Evaluation of subcooled critical heat flux correlations using the PU-BTPFL CHF database for vertical upflow of water in a uniformly heated round tube. Nucl. Tech. 117, 234–247.

Hewitt, G.F., 1982. Burnout. In: Hetsroni, G. (Ed.), Handbook of Multiphase Systems, vol. 1. Hemisphere Publishing.

Iwamura, T., et al., 1992. Evaluation of mechanistic DNB models using HCLWR CHF data, JAERI-M 92-033, JAERI.

Katto, Y., 1992. A prediction model of subcooled water flow boiling CHF for pressure in the range 0.1–20 MPa. Int. J. Heat Mass Transfer 35, 1115–1123.

Katto, Y., 1994. Critical heat flux. Int. J. Multiphase Flow 20, 53–90.

Kirby, G.J., Stainiforth, R., Kinneir, L.H., 1967. A visual study of forced convection boiling-2. Flow patterns and burnout for a round test section, AEEW-R-506, UKAEA, Winfrith.

Kutateladze, S.S., 1969. Problems of Heat Transfer and Hydraulics of Two-phase Media, Academy of Science, Moscow (O.M. Blunn, J.G. Collier, Trans.). Pergamon, London.

Lahey, R.T. Jr., Moody, F.J., 1993. The Thermal-Hydraulics of a Boiling Water Nuclear Reactor, 2. American Nuclear Society, La Grange Park, IL.

Lee, C.H., Mudawar, I., 1988. A mechanistic critical heat flux model for subcooled flow boiling based on local bulk flow conditions. Int. J. Multiphase Flow 14, 714–728.

Lee, K.W., et al., 1997. An improved and simplified DNB model based on mass, energy and momentum balance equations. In: Proceedings of the Fifth International Conference on Nuclear Thermal Hydraulics, Operations and Safety, Beijing, China, 1997.

Lee, S.C., Dorra, H., Bankoff, S.G., 1992. A critical review of predictive models for the onset of significant void in forced-convection subcooled boiling, HTD, Fundamentals of Subcooled Flow Boiling, ASME, 217, 33–39.

Levy, S., 1967. Forced convection subcooled boiling-prediction of vapor volumetric fraction. Int. J. Heat Mass Transfer 10, 951–965.

Mayinger, F., 1967. Investigation into the critical heat flux in boiling water, EUR-3347.

Merilo, M., Ahmad, S.Y., 1979. The effect of diameter on vertical and horizontal flow boiling crisis in a tube cooled by Freon-12, AECL-6485, Chalk River Nuclear Laboratories.

Moon, S.K., Baek, W.P., Chang, S.H., 1996. Parametric trends analysis of the critical heat flux based on artificial neural networks. Nucl. Eng. Des. 163, 29–49.

Nariai, H., Inasaka, F., 1992. Critical heat flux and flow characteristics of subcooled flow boiling with water in narrow tubes. In: Jones, O.C., Michiyoshi, I. (Eds.), Dynamics of Two-phase Flows. CRC Press, Boca Raton, FL, pp. 689–708.

Nedderman, R.M., Shearer, C.J., 1964. Correlation for the friction factor and velocity profile in the transition region for flow in sand-roughened pipes. Chem. Eng. Sci. 19, 423–428.

Saha, P., Zuber, N., 1974. Point of net vapor generation and vapor void fraction in subcooled boiling. In: Proceedings of the Fifth International Heat Transfer Conference, Tokyo, Japan, vol. IV, pp. 175–179.

Staub, F.W., 1968. The void fraction in subcooled boiling-prediction of the initial point of net vapor generation. J. Heat Transfer 90, 151–157.

Stevens, J.R., Miles, D.N., 1980. Boiling crisis data for vertical upflow of Freon-12 in round tubes and annular channels, AAEC/E506, Australian Atomic Energy Commission.

Styrikovitch, O., Nevstrueva, E.I., Dvorina, G.M., 1970. The effect of two phase flow pattern on the nature of heat transfer crisis in boiling. In: Proceedings of the Fourth International Heat Transfer Conference, Paris-Versailles, vol. VI, B 6.10.

Taylor, R.P., Coleman, H.W., Hodge, B.K., 1985. Prediction of turbulent rough-wall skin friction using a discrete element approach. J. Fluid Engineering 107, 251–257.

Taylor, R.P., Hodge, B.K., 1992. Validated heat-transfer and pressure-drop prediction methods based on the discrete element method: Phase I, Three-dimensional roughness, ANL/ESD/TM-31, Argonne National Laboratory.

Thompson, B., Macbeth R.V., 1964. Boiling water heat transfer in uniformly heated round tubes: A compilation of world data with accurate correlations, AEEW-R-356.

Tong, L.S., 1964. DNB (Burnout) studies on an open lattice core, Westinghouse report WCAP-3736.

Tong, L.S., Hewitt, G.F., 1972. Overall view point of flow boiling CHF mechanisms, ASME Paper 72-HT-54.

Tong, L.S., Weisman, J., 1996. Thermal Analysis of Pressurized Water Reactors, 3. American Nuclear Society, La Grange Park, IL.

Weatherhead, R.J., 1963. Nucleate boiling characteristics and the critical heat flux occurrence in sub-cooled axial flow water system, ANL-6675.

Weisman, J., Pei, B.S., 1983. Prediction of critical heat flux in flow boiling at low qualities. Int. J. Heat Mass Transfer 26, 1463–1477.

Weisman, J., Ileslamlou, S., 1988. A phenomenological model for prediction of critical heat flux under highly subcooled conditions. Fusion Tech. 13, 654–659.

Weisman, J., 1992. The current status of theoretically based approaches to the prediction of the critical heat flux in flow boiling. Nucl. Tech. 99, 1–21.

Ying, S.H., Weisman, J., 1986. Prediction of the critical heat flux in flow boiling at intermediate qualities. Int. J. Heat Mass Transfer 29, 1639–1648.

Zenkevich, A., 1974 Analysis and generalization of experimental data on heat transfer crisis associated with forced convection of cooling water in tubes, AECL-Tr-Misc-304.

Zuber, N., Findlay, J., 1965. Average volumetric concentration in two-phase flow system. J. Heat Transfer 87, 453–462.